

CIS 301: Logical Foundations of Programming

Spring 2026

Exam 2 – 100 points

This test is closed-notes and closed-computers.

This exam has 10 multiple choice and 4 open-ended problems.

Name: _____ Solution _____

Score: _____

Multiple choice – 10 questions, 3 points each.

Clearly circle your answer or write the letter of your answer next to each question.

1. What is the reasoning behind the Implyl rule?
 - a) If we can conclude statement a whenever statement b is true, then it follows that $a \rightarrow b$
 - b) If we can conclude statement b whenever statement a is true, then it follows that $a \rightarrow b$**
 - c) If we know $a \rightarrow b$ and a are both true, then it follows that b is also true
 - d) If we know $a \rightarrow b$ and b are both true, then it follows that a is also true

2. What statement contradicts with $a \rightarrow \neg b$?
 - a) $\neg(a \rightarrow \neg b)$**
 - b) $b \rightarrow \neg a$
 - c) $\neg(a \wedge b)$
 - d) None of the above

3. Is the following predicate logic statement over the set of integers ($\mathbb{Z}$) true or false?
$$\neg(\exists x (x > 0 \wedge (x-2)^2 < 0))$$
 - a) True, because the square of any integer will never be less than 0**
 - b) True, because when $x = 3$, then $(3-2)^2 = 1$, which is not less than 0
 - c) False, because the square of any integer will never be less than 0
 - d) False, because when $x = 3$, then $(3-2)^2 = 1$, which is not less than 0

4. Consider the sets $A = \{2,5,8,9,12,13\}$ and $B = \{1,2,3,5,7,8,10,12\}$. Find $B-A$.
 - a) $\{2,5,8,12\}$
 - b) $\{1,3,7,10\}$**
 - c) $\{9,13\}$
 - d) $\{1,3,7,9,13\}$

5. What is the negation of the following predicate logic statement? (It doesn't matter what the predicates are for the purposes of the problem, but you're welcome to invent some.)

$$\exists x \exists y (P(x) \wedge \neg Q(x, y))$$

- a) $\forall x \forall y (\neg P(x) \vee \neg Q(x, y))$
- b) $\forall x \forall y (\neg P(x) \wedge \neg Q(x, y))$
- c) $\forall x \forall y (P(x) \rightarrow Q(x, y))$**
- d) $\forall x \forall y (\neg P(x) \rightarrow Q(x, y))$

6. Is the following predicate logic statement over the set of integers ($\mathbb{Z}$) true or false?

$$\neg(\forall x \forall y (x > y \wedge x > 2 \rightarrow x - y > 2))$$

- a) True, because when $x = 3$ and $y = 2$, $3 - 2$ is not greater than 2**
- b) True, because when $x = 4$ and $y = 1$, $4 - 1$ is greater than 2
- c) False, because when $x = 3$ and $y = 2$, $3 - 2$ is not greater than 2
- d) False, because when $x = 4$ and $y = 1$, $4 - 1$ is greater than 2

7. Consider the sets $A = \{1, 4, 7, 9, 12, 14, 15\}$, $B = \{4, 6, 7, 8, 11, 12, 14\}$, and $C = \{1, 2, 7, 8, 9, 12, 13, 15, 16\}$. Find $A \cup (B \cap C)$.

- a) $\{1, 4, 7, 8, 9, 12, 14, 15\}$**
- b) $\{1, 7, 8, 12, 15\}$
- c) $\{1, 4, 6, 7, 9, 11, 12, 14, 15\}$
- d) None of the above

8. Is natural deduction in propositional logic *complete*?

- a) Yes, because we can use natural deduction to prove every argument that is valid in a truth table**
- b) Yes, because every argument we can prove with natural deduction is also valid in a truth table
- c) No, because there are some arguments that are valid in a truth table that we can't prove using natural deduction
- d) No, because there are some arguments that we can prove using natural deduction that are not valid in a truth table

9. Suppose we have three predicates over the domain of household items:

$K(x)$: item x is in the kitchen

$F(x)$: item x is a fork

$D(x)$: item x is dirty

What is the best interpretation of the following predicate logic statement?

$$\forall x \neg(F(x) \wedge D(x) \rightarrow K(x))$$

- a) All forks that are not dirty are in the kitchen
- b) All items that not dirty forks are in the kitchen
- c) No dirty forks are in the kitchen**
- d) Not all dirty forks are in the kitchen

10. Bob claims that for sets A , B , and C , if $A \neq B$, then it must be the case that $A \cap B \neq A \cap C$. Is Bob right or wrong, and why?

- a) Bob is right, because for $A = \{1,2,3\}$, $B = \{1\}$, $C = \{2\}$, then $A \neq B$ and $A \cap B \neq A \cap C$.
- b) Bob is right, because if two sets don't have the same elements, then they can't have the same elements in common with another set.
- c) Bob is wrong, because for $A = \{1,2,3\}$, $B = \{1\}$, $C = \{2\}$, then $A \neq B$ and $A \cap B \neq A \cap C$.
- d) Bob is wrong, because for $A = \{1,2,3\}$, $B = \{4\}$, $C = \{5\}$, then $A \neq B$ and $A \cap B = A \cap C$.**

Open Ended – 4 questions, 16-20 points each.

11. (17 pts) Consider the following questions about sets.

a) (3 pts) List the elements in the set:

$$\{x : x = 4k+7, k \in \mathbb{Z}, -3 \leq k \leq 2\}$$

$\{-5, -1, 3, 7, 11, 15\}$

b) (3 pts) List the elements in the set:

$$\{x : 3x = 2k+1, x \in \mathbb{Z}, k \in \mathbb{Z}, 0 \leq k \leq 8\}$$

$\{1, 3, 5\}$

c) (3 pts) List the elements in the set:

$$\{x : x = 3p - 4q, p \in \mathbb{Z}, q \in \mathbb{Z}, 5 \leq p \leq 7, 1 \leq q \leq 2\}$$

$\{7, 10, 11, 13, 14, 17\}$

d) (4 pts) Use set builder notation to describe a set with the elements below. Use only mathematical notation and not words to describe the set.

$$\{-5, 1, 7, 13, 19, 25, 31\}$$

$$\mathbf{\{x : x = 6k + 1, k \in \mathbb{Z}, -1 \leq k \leq 5\}}$$

e) (4 pts) Use set builder notation to describe a set with the elements below. Use only mathematical notation and not words to describe the set (hint: think about powers of 3).

$$\{3, 9, 27, 81\}$$

$$\mathbf{\{x : x = 3^k, k \in \mathbb{N}, k \leq 4\}}$$

12. (20 pts) Complete the following natural deduction proof.

$(a \rightarrow d \wedge f, d \wedge c \rightarrow \neg e) \vdash (a \wedge b \rightarrow \neg(c \wedge e))$

Proof(

1 ($a \rightarrow d \wedge f$) by Premise,

2 ($d \wedge c \rightarrow \neg e$) by Premise,

//FINISH THE PROOF BELOW

3 SubProof(

4 Assume ($a \wedge b$),

5 (a) by AndE1(4),

6 (b) by AndE2(4),

7 ($d \wedge f$) by ImplyE(1, 5),

8 (d) by AndE1(7),

9 (f) by AndE2(7)

10 SubProof(

11 Assume ($c \wedge e$),

12 (c) by AndE1(11),

13 (e) by AndE2(11),

14 ($d \wedge c$) by AndI(8, 12),

15 ($\neg e$) by ImplyE(2, 14),

16 ($\bot$) by NegE(13, 15)

),

17 ($\neg(c \wedge e)$) by NegI(10)

),

18 ($a \wedge b \rightarrow \neg(c \wedge e)$) by ImplyI(3)

)

13. (16 pts) Suppose we have the following predicates over the domain of houses:

TwoStory(x) – house x has two stories

Basement(x) – house x has a basement

Fence(x) – house x has a fenced backyard

Bigger(x, y) – house x is bigger than house y

New(x) – house x is new

Translate the following English statements to predicate logic.

a) (4 pts) There is a new two story house with a basement.

$\exists x (\text{New}(x) \wedge \text{TwoStory}(x) \wedge \text{Basement}(x))$

b) (4 pts) Not all new houses have fenced backyards.

$\neg(\forall x (\text{New}(x) \rightarrow \text{Fence}(x)))$

c) (4 pts) There is a new house with a basement that is bigger than every two story house.

$\exists x \forall y (\text{New}(x) \wedge \text{Basement}(x) \wedge \text{TwoStory}(y) \rightarrow \text{Bigger}(x, y))$

d) (4 pts) No two story house has both a basement and a fenced backyard.

$\neg (\exists x (\text{TwoStory}(x) \wedge \text{Basement}(x) \wedge \text{Fence}(x)))$

14. (17 pts) Next to each “??”, write the appropriate claim, assumption, or justification to complete the natural deduction proof.

($\forall ((x: T) \Rightarrow (\neg P(x) \rightarrow Q(x)))$)

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($\forall ((x: T) \Rightarrow (\neg Q(x) \rightarrow P(x)))$)

Proof(

1 ($\forall ((x: T) \Rightarrow (\neg P(x) \rightarrow Q(x)))$) by Premise,

2 Let ((a: T) => SubProof(

3 SubProof(

4 Assume($\neg Q(a)$),

5 ($\neg P(a) \rightarrow Q(a)$) by $\text{AllE}[T](1)$,

6 SubProof(

7 Assume ($\neg P(a)$),

8 ($Q(a)$) by $\text{ImplyE}(5, 7)$,

9 (F) by $\text{NegE}(8, 4)$

),

10 ($P(a)$) by $\text{PbC}(6)$

),

11 ($\neg Q(a) \rightarrow P(a)$) by $\text{ImplyI}(3)$

)),

12 ($\forall ((x: T) \Rightarrow (\neg Q(x) \rightarrow P(x)))$) by $\text{AllI}[T](2)$

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